

Alexander Filak



About me

I am a highly motivated first-year Master's student at the Moscow Institute of Physics and Technology (MIPT), specializing in Artificial Intelligence at the Center for Information, Telecommunication, and Information Systems (CITIS). I am proud to be part of the Phystech School of Radio Engineering and Computer Technology (DREC), which provides me with a strong foundation in cutting-edge technical education and research opportunities.

personal

Alexander Filak
23 years

Areas of specialization

CV · ML

Interests

Computer Science
Systems Programming
Theoretical Physics
Mathematical Physics

 @n0t_KNOW

 filaka771

EDUCATION

2020–2025 **MIPT**
BACHELOR · DREC



2025–Now **MIPT**
MASTER · DREC



EXPERIENCE

2022 – Now

Tutor in higher mathematics and theoretical physics. Worked with students from MIPT, HSE, MAI, etc.

2023 – Now

Mathematics teacher at the Summer Multidisciplinary School. Author of courses on Group Theory, Differential Equations, etc.

PROGRAMMING



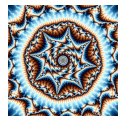
PROJECTS

Paracl

Compiler for a simple C-like language in C++, with a hand-written lexer, recursive-descent parser, x86-64 code generation, and a small syscall-based runtime for integer I/O, targeting FASM assembly.

Mandelbrot Set

Fractal renderer in C++ featuring multiple Mandelbrot implementations: a scalar CPU version, an AVX2-optimized renderer, a Boost.Multiprecision/MPFR deep-zoom path, and custom PPM image generation with configurable shader-based coloring.



CPU

Virtual CPU in C featuring a full toolchain with a two-pass assembler, binary disassembler, and register/stack-based emulator supporting arithmetic, branching, function calls and indirect addressing.

LRU Cache

Generic LRU cache implementation in C++ designed for efficient recency tracking and eviction, with libFuzzer-based randomized testing against a naive vector-based implementation.

LLVM Contribution

Merged upstream LLVM/Clang contribution fixing alignment in `-analyzer-help` output via `AnalyzerOptions::printFormattedEntry`, with added unit tests covering formatting edge cases in analyzer option listing.

LANGUAGES

Language	Level
Russian	C2
English	B2

mother tongue
● ● ● ●

PUBLICATIONS

2025 Filak, A. Yu., & Mozolina, N. V. (2025). Study of the applicability of general-purpose hash functions in the problem of data integrity monitoring and deduplication. *Voprosy zashchity informatsii [Information Security Issues]*, 2025(1). EDN: HZQJSI. DOI: 10.52190/2073-2600_2025_1_8.

Alexander Filak  filaka@phystech.edu

 8 (916) 558 35 66  @n0t_KNOW

 filaka771